MIL-G-26762B (USAF)
09 May 1988

SUPERSEDING
MIL-G-26762A (USAF)
2 September 1970

MILITARY SPECIFICATION

GAGES, DIAL BORE

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers several types, styles, and classes of dial bore gages normally used by the Air Force.

1.2 Classification.

1.2.1 Types. Gages shall be of the following types, styles and classes as specified. (See 6.2).

Type I Plug or Disc

Style - 1 Regular or Vertical
Style - 2 Pistol grip (with trigger)

Type II Centralizing Plungers, or 2 point contact single centralizing plunger.

Style - 1 Regular or Vertical
Style - 2 Regular or Vertical (with pistol grip attachment)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMIRFW, Robins AFB, GA. 31098-5609 by using the self-addressed Standardization Document improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Type III Retracting Contacts

Style - 1	Regular or Vertical (with push button)
Style - 2	Pistol grip (with trigger)

Type IV Pin

Style - 3	In-line (Dial bore gage assembly)

Indicating Dial Graduations

Class - A*	0.0001 inch increments
Class - B*	0.0005 inch increments
Class - C*	0.001 inch increments

*(The Class A, B, and C Indicating Dial Graduations are equivalent to types I, II, and III indicators as specified in Commercial Item Description A-A-2348.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and Standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

PPP-P-40	Packaging and packing of hand tools

STANDARDS

Federal

FED-STD-H28	Screw-thread standard for federal services

Military

MIL-STD-105	Sampling procedure and tables of inspection by attributes
MIL-STD-129	Marking for shipment and storage
MIL-STD-130	Identification marking of V.5. military property

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2.1.2 Other Government Documents and Publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

COMMERCIAL ITEM DESCRIPTION

A-A-2348 Indicators, dial accessories and test sets

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.6 and 6.4).

3.2 Design. The dial bore gages shall be designed for positive centralization with either centralizating plungers, discs, pins or centralizer (spherically radiused, single plunger type). Design of the gage shall include easy means of interchanging components, setting to masters, and gaging of hole sizer as specified herein.

3.2.1 Interchangeability. All parts of the gage shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or the gage.

3.3 Construction. The gage shall be constructed of materials that will withstand normal shop usage. Construction features shall include easy means of assembling and adjusting various components to cover a range of hole sizes with a single gage (set) within its range as shown in table I.

3.3.1 Welding, Brazing or Soldering. Welding, brazing, or soldering shall be employed only where those operations are required in the original design. These operations shall not be employed as repair measures for defective parts.

3.3.2 Fastening Devices. All screws, pins, bolts, and similar fasteners shall be installed with means for preventing loss of tightness. Fasteners subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently deformed.

3.3.3 Surfaces. All surfaces of castings, forginge, molded parts, stampings and welded parts shall be clean and free from sand, dirt, fins, sprues, scale, flux and other harmful or extraneous materials. External surfaces shall be smooth, and all edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. There shall be no cadmium plating or any exterior surfaces of the assembled gage.

3.3.4 Painting. Unless otherwise specified (see 6.2), the gage shall be painted in accordance with the manufacturer's standard commercial practice.

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3.4 Components. Components of the basic gage assembly shall consist essentially of a gaging head assembly, linkage tube or body, and dial gage with housing or pistol grip.

3.4.1 Gaging Head Assemblies. Gaging head assemblies shall be of the following type as specified in 1.2:

a. Type I. Type I gaging head assembly shall consist of various diameters of discs or be of the interchangeable plug type. The disc or plug shall serve as backup, diametrically opposite, contact to dial actuating contact. The disc or plug shall also serve as hole (gaged) centering means.

b. Type II. Type II gaging head assembly shall consist of two spring loaded centralizing plungers or be spherically radiused single plunger which contacts the circumference of the gaged hole at two points. The dial gage actuating plunger shall be centered between the centralizing contacts and be diametrically opposite the reference, adjustable, backup, contact.

c. Type III. Type III gaging head assembly shall consist of three point, retracting type, with interchangeable, contacts. The three point contacts shall provide gaging of hole size and positive centralization with simultaneous expanding, equalized contacts. When specified (see 6.2) two point contacts shall be furnished in lieu of three point type.

d. Type IV. Type IV gaging head assembly shall consist of a dial actuating contact point at one end (dial unit end) and, in line, reference, adjustable, back up contact point at opposite end (spacer or tube end).

3.4.2 Range Extensions (I.D.). All types of dial bore gages, except type I shall have a threaded, adjustable, range extension (reference or backup contact) complete with lock nut.

3.4.3 Dial Indicator Actuating and Centralizing Contacts. The dial indicator actuating and centralizing contacts shall operate freely, with means for preventing loss of the contacts from the gaging head assembly or its housing.

3.4.4 Contacts, Material and Hardness. Unless otherwise specified (see 6.2), the contacts shall be of alloy steel, hardened to not less than Rockwell C-55. When specified (see 6.2), the contacts shall be equipped with carbide or diamond tips. The tips of the contacts shall be rounded and polished to a 10 microinch or better finish.

3.4.5 Range Extension (Length or Depth) A range extension tube shall be provided to obtain the length or depth of gaging in hole. The extension tube shall be of steel or aluminum with easy means of clamping the gaging head assembly in 3.4.1 and the dial gage housing in 3.4.6 to form a single unit construction (complete gage assembly). An extension rod of alloy steel, hardened to not less than Rockwell C-55 at contact and bearing surface areas, will be provided in the extension tube for actuating dial gage, as applicable to type gage on contract or order.

3.4.6 Dial Indicator Housing. The dial indicator housing, with the exception of Type-I, Style-1, shall be either a casting of aluminum or molded of plastic. The housing shall protect the dial indicator from damage in normal shop use.

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3.4.7 Dial Bore Gage Styles.Dial bore gage styles shall be as follows and as specified in 1.2:

a. Style 1. Style 1 regular or vertical dial bore gage shall be an in-line dial and extension tube or body (length or depth) with gaging head assembly at right angle to dial gage and extension body, as specified in 1.2.

b. Style 2. Style 2 shall be a pistol grip style handle of cast aluminum or molded plastic. When specified (see 6.2), the pistol grip style shall be furnished with the Type I plug, Type II and III dial bore gages. The Type I plug, and Type III shall include a trigger for retracting the gaging contacts.

c. Style 3. Style 3 shall be an in-line bore gage assembly having indicator contact at one end and reference contact at the opposite end of the assembly. The Style 3 gage is applicable to the Type IV pin as specified in 1.2.

3.5 Case. The gage assembly and its components shall be arranged in a wood, metal or plastic case, with a hinged or sliding cover. The gage components shall be arranged (cradled or separators) in a manner that they will be protected from damage in shipment and normal shop use.

3.6 Dial Indicator. The dial indicator used with the dial bore gage shall conform to Commercial Item Description A-A-2348 and as specified in 1.2. Unless otherwise specified (see 6.2) dial indicators shall be of the balanced dial type, be adjustable, and when applicable, have revolution counter.

3.7 Sizes and Capacities. Sizes and capacities of the dial bore gage shall be as specified in Table I. Requirements shall be not less than those specified, except where a range is shown - the low or minimum shall be not more than; the high or maximum shall be not less than.

3.8 Equipment.

3.8.1 Tools and Accessories. The following tools and accessories shall be furnished with, and adaptable to each dial bore gage set as specified herein:

a. Wrenches normally furnished by the supplier for use with the gages.

b. Complete set of dial bore gage components consisting of various sizes of centering disc, centralizing plungers, interchangeable plugs, interchangeable extensions, indicator contact points, dial indicator, reference or adjustable (range) contacts, dial housing and spacers (dial length of bore), as applicable to the gage sets specified herein, and to cover the range as normally furnished by the supplier.

3.9 Nameplate. Unless otherwise specified (see 6.2), a nameplate (decal, ink stamped, imbossed or equal) shall be attached to case of each dial bore gage set. The nameplate shall conform to MIL-STD-130 and contain the following data:

Nomenclature
Manufacturer's Name
Manufacturer's Model Number
Federal Stock Number
U S

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TABLE I. SIZES AND CAPACITIES

SIZE (as specified in 6.2).	HOLE SIZE RANGE (I. D. Inches)	GAGING, DEPTH INCHES, MIN. Unless otherwise specified (see 6.2).
TYPE I GAGE:		
1. (Style 1)	.078 - .266	13/16
2. (Style 1)	.122 - .250	2 1/4
TYPE II GAGE:		
3. (Style 1)	.217 - .594	1 1/2
4. (Style 1)	.560 - 1.565	2 5/16
5. (Style 2)	.187 - 1.000	2 3/4 - 4 3/4 (Range)
6.	.250 - .500	2 1/4
7.	.500 - .750	3
8.	.750 - 1.250	3
9.	1.250 - 2.250	3
10.	2.250 - 4.000	6
11.	4.000 - 6.000	6
12.	6.000 - 12.125	6
(OPTIONAL SIZES)		
13.	1.500 - 3.000	3 1/8
14.	3.000 - 5.187	6
15.	5.000 - 8.125	6
16.	8.000 - 12.125	6
TYPE III GAGES:		
17.	.670 - 1.250	3
18.	1.250 - 2.750	3
19.	2.750 - 6.250	6
TYPE IV GAGE:		
20.	4.00 - 40.00	

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for Compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of Inspections. Inspections shall be classified as follows:

a. Production inspection (see 4.3).

b. Inspection of preparation for delivery (see 4.5).

c. First active inspection (see 4.6).

4.3 Production Inspection. Each dial bore gage to be delivered shall undergo all examinations specified in 4.4 and all tests specified in 4.5 through 4.5.2. None of the items that fail to be acceptable under these examination and test requirements shall be delivered under this contract.

4.4 Examination. The gage sets shall be examined for design, construction, materials, components, and workmanship to determine compliance with the requirements of this specification. Any deviation from these requirements shall be cause for rejection.

4.4.1 Inspection Conditions. Unless otherwise specified, all inspections shall be performed at a temperature of between 70° and 80°F (21.1° to 26.7°C).

4.5 Tests.

4.5.1 Operational or Performance Test. The operational or performance test shall consist of setting the dial bore gage to a master ring gage and gaging of the hole at least 25 times. During this period of hole gaging, all components of the gage shall be observed for proper function of all components to provide requirements specified herein.

4.5.2 Dial Bore Gage (Set) Test. The dial bore gage should be set to zero using a master micrometer, gage blocks, or a ring gage, at the minimum range of the dial bore gage.The accuracy of the master micrometer, gage block or ring gage shall be at least twice the accuracy of the bore gage tested. The master micrometer and gage blocks cannot be used to set a Type III (3 point contact) bore gage. The dial bore gage (all types) shall be checked for repeatability by inserting the dial bore gage into each different size ring gage. The dial bore gage (all types) shall be verified for accuracy of gaging bores by using master ring gages at medium and maximum range of the dial bore gage. At least 3 gagings shall be made of each range to determine compliance with the following accuracies and tolerances:

a. Repeatability. Repeatability tolerance (3 readings each range) shall be as follows:

(1) Class A + or - 0.00005 inch

(2) Class B + or - 0.00025 inch

(3) Class C + or - 0.0005 inch

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b. The accuracy of the measurements taken on the medium and maximum range ring gage referenced to the minimum ring gage shall be as follows:

(1) Class A + or - 0.0001 inch

(2) Class B + or - 0.0005 inch

(3) Class C + or - 0.001 inch

Shift gage in hole for side pressure left and right and gage shall move immediately to minus reading each shift (left and right). Plus reading indicates centralizer(s) are not centered with gaging contacts and is not acceptable.

4.6 First Article Inspection. When a first article is required, the first article shall undergo all examinations specified in 4.4 and all tests specified in 4.5 through 4.5.2. If the item fails to meet these criteria, it shall not be delivered under this contract.

4.7 Inspection of Preparation for Delivery. Preservation, packaging, packing and marking for shipment and storage shall be inspected to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY.

5.1 Packaging.

5.2 Preservation, Packaging, Packing and Marking. Unless otherwise specified (see 6.2), preservation, packaging, packing and marking shall be in accordance with PPP-P-40. The levels of preservation, packaging, and packing shall be as specified (see 6.2).

6. NOTES.

6.1 Intended Use. The gages covered in this specification are intended for use in aircraft shops, i.e., engine overhaul, machine, inspection and etc. They are essentially designed and used for determining bore sizes, but may also determine out of round, bell mouth and tapered conditions of holes.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Type, style, class and size gages required (see 1.2 and TABLE - I).

c. Painting, if different (see 3.3.4).

d. Two point contact type, if required (see 3.4.1(c)).

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e. Dial indicator shall be of the continuous dial type, if required (see 3.6).

f. Gage depth or length, if different (see TABLE I).

g. Nameplate, if different (see 3.9).

h. Preservation, packaging, packing, marking and levels, if different and as specified (see 5.1).

6.3 Subject Term (Key Word) Listing:

Bow gage
Plug
Disc
Centralizing
Retracting
Pin

6.4 First Article. When a first article inspection is required the item should be a first article sample. The first article shall consist of one unit. The contracting officer shall include specific instructions in aquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitation for bids should provide that the government deserves the right to waive the requirement for samples for first article inspections to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Changes from Previous Issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Table II. Part Numbers Based on Type, Style and Class.

PART NO.	TYPE	STYLE	CLASS
M26762/1-1	I	1	A
M26762/1-2	I	1	B
M26762/1-3	I	1	C
M26762/2-1	I	2	A
M26762/2-2	I	2	B
M26762/2-3	I	2	C
M26762/3-1	II	1	A
M26762/3-2	II	1	B
M26762/3-3	II	1	C
M26762/3-1	II	2	A
M26762/3-2	II	2	B
M26762/3-3	II	2	C
M26762/4-1	III	1	A
M26762/4-2	III	1	B
M26762/4-3	III	1	C
M26762/4-1	III	2	A
M26762/4-2	III	2	B
M26762/4-3	III	2	C
M26762/5-1	IV	3	A
M26762/5-2	IV	3	B
M26762/5-3	IV	3	C

CUSTODIAN:

AIR FORCE - 99

PREPARING ACTIVITY:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-26762B (USAF)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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